Justin Vinall

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Senior backend engineer (Python, SQL, Azure) with 15 years in software, shipping services and data pipelines that stay fast and reliable in production, and rearchitecting the ones that aren't. Also a decade of C#/.NET; builds daily with AI-assisted development and LLM APIs.

EXPERIENCE

Avanade, Seattle, WA — *Senior Backend Developer*

JAN 2018 – PRESENT

Accenture and Microsoft's joint venture. Lead backend engineer on client engagements.

Hydraulic Fracturing Analytics — Fortune 100 energy company · 2025–Present
Python, Azure Databricks, SQL Server, GIS

- Own the daily Azure Databricks pipeline that joins multiple source tables to map thousands of wells into geospatial analytics and Spotfire layers
- Built simulation and optimization logic with petroleum engineers to estimate fracture length by stage from fluid volume, permeability, and location
- Generated and validated fracture polygons with Shapely, GeoPandas, and alphashape, quantifying overlap and identifying under-fractured areas
- Designed SQL Server schemas and data access for the spatial outputs in Spotfire

Agent Assist — Fortune 500 utility contact center · 2022–2023
Python, Azure Functions, SQL Server

- Cut average response time from 30 seconds to 3 seconds by rearchitecting the Python Azure Functions behind 17 parallel APIs used on ~10,000 calls a day
- Led migration from SAP HANA to SQL Server, translating and validating nearly 5,000 lines of SQL and designing indexes for the new schema
- Load-tested the migrated database and queries at storm peaks, 5x normal traffic
- Raised Python test coverage above 90% and built Azure DevOps CI/CD pipelines
- Monitored Application Insights daily and investigated every production error

Microsoft — Partner Network platform · 2018–2021
C#, ASP.NET MVC, Azure Durable Functions, Service Bus, JavaScript, Sitecore

- Built an Azure Durable Functions workflow using Service Bus to migrate customer data between platforms via application APIs
- Integrated Marketo REST APIs and forms to capture customer data for analytics
- Built full-stack features and reusable Sitecore components in C# and JavaScript

Vertafore, Bothell, WA — *Software Engineer II, Test Automation*

SEP 2011 – DEC 2017

- Led 5 offshore engineers building C# and Selenium WebDriver UI automation for ~300 flows of a new ASP.NET insurance application, reviewing all their code
- Developed customer database migration and performance-testing tools, and ran smoke testing and live troubleshooting for every production release

PROJECTS

JobScan — personal LLM pipeline that ranks job postings for review
Python, Anthropic API, SQLite, pytest · github.com/pirsqua/JobScan

- Designed a two-stage LLM evaluation that has screened 1,600+ postings: a low-cost model triages and a larger one scores the rest against a rubric
- Added code-side guardrails after measuring model failure modes: skips must quote the posting, and borderline results get a second evaluation
- Integrated 17 job-board platforms with concurrent fetching and validated each model upgrade against known postings first; 500+ tests

SKILLS

Languages: Python, SQL, C#

Databases: SQL Server, T-SQL, PostgreSQL, MySQL, Cosmos DB

Python & Data: PySpark, Azure Databricks

APIs & Integration: REST, OpenAPI/Swagger, third-party APIs, Azure API Management, Azure Service Bus

Microsoft Azure: Functions, App Service, Entra ID, Key Vault, Storage, VNet, Application Insights

DevOps: Azure Pipelines, ARM templates, Bicep, Docker

Testing: unit, integration, load, performance

AI & LLM: Claude Code, Anthropic API, structured outputs, prompt caching

.NET: ASP.NET Core, ASP.NET MVC/Web API

CERTIFICATIONS

Microsoft Certified: Azure Developer Associate

EDUCATION

Western Washington University
B.S. in Computer Science, Minor in Mathematics

MAR 2011